

Yizheng Jiao

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RESEARCH INTERESTS

My research interests focus on the design and implementation of efficient and secure data management system. I am doing research on write-optimized index data structures for high-speed storage system. In my research projects, I am developing a general-purpose in-kernel file system on Linux which has compelling performance on micro-write intensive workloads. I am also exploring the potentials of write-optimized data structures in designing secure persistent key-value stores with the aid of SGX/TrustZone to protect user's data against malicious Operating System.

EDUCATION EXPERIENCE

Ph.D in Computer Science Aug. 2016 - Present
University of North Carolina at Chapel Hill
Advisor: Donald Porter Expected graduation date: Aug. 2021

M.S. in Computer Science Jan. 2014 - Aug. 2016
Stony Brook University
Advisor: Donald Porter

M.S. in Electrical and Computer Engineering Sep. 2009 - March. 2012
Huazhong University of Science & Technology
Advisor: Feng Bin, Wenyu Liu

B.S. in Electrical and Computer Engineering Sep. 2005 - Jun. 2009
Chongqing University

PROFESSIONAL EXPERIENCE

Research Intern May 2018 - Aug. 2018
Advanced Technology Group, NetApp Inc, Durham, N.C.
Mentor: Michael Condict, Xiongzi Ge

- Optimization of *BetrFS* for SSD device;
- Integration of in-kernel B^e -tree key-value store with Linux page cache to reduce overhead of data transport;
- Extended *BetrFS* to support large block size to reduce meta-data overhead;
- Sequential write performance is increased by 2 times.

Software Engineer Intern May 2017 - Aug. 2017
vSan Team, VMWare Inc, Palo Alto, C.A.
Mentor: Wenguang Wang, Ding Li

- Optimization of Copy-on-Write B -tree with write-optimized techniques;
- Analysis of snapshot service performance for Virtual Distributed File System (VDFS);
- Implementation of physical block cache for VDFS;
- Read performance is improved by 50%.

Software Engineer May 2013 - Aug. 2013
Leadership Computing Facility Group, Oak Ridge National Lab, Knoxville, T.N.
Mentor: Sarp Oral

- Implementation of parallel and scalable programs to move and archive large volumes of data on supercomputers;
- Analysis on performance of implemented utilities on Titan supercomputer with hundreds of MPI nodes.

RESEARCH PROJECTS

BetrFS Project

Jan. 2014 - Present

Details could be found at <http://www.betrfs.org>

- Profing and analyzing BetrFS performance on SSD device;
- Optimizing BetrFS's metadata performance with varied block size;
- Integrating in-kernel B^c -tree key-value store with Linux page cache;
- Implementing Copy-on-Write page cache to support file system transaction.

Exploring Contiguity in Memory System

May 2014 - May 2015

- Studying physical page allocator of popular operating systems, such as Linux, Windows, OpenBSD and Solaris OS;
- Studying disk block allocator of file systems , such as Ext4, ZFS;
- Evaluating performance impact of Linux Transparent Hugepages for application with big memory footprint, such as graph proccessing, k-mer counting for Genome sequences.

Efficient Data Management for MapReduce Programs

May 2013 - Dec 2013

- Enabling memory sharing between map/reduce tasks on the same compute node;
- Accelerating merge phase of MapReduce programs by decoupling values from the K-V pairs during sorting;
- Testing performance of MapReduce programs on Amazon EC2 cluster.

Hybrid Memory System for GPU

Aug. 2012 - May 2013

- Hybridizing phrase-change memory and DRAM for GPU global memory;
- Implementing compiler-directed initial data placement for CUDA programs;
- Integrating GPGPU-Sim with DRAMSim2 to simulate GPU global memory system;
- Implementing simulator code to perform hardware-based page migration between PCM and DRAM.

PUBLICATIONS [Arxiv]: Yizheng Jiao. *Exploring Physical Contiguity in Memory System*.

[Submitted to ATC'2021]: Yizheng Jiao, Simon Bertron, Nirjhar Mukherjee, Rory Bennett, Michael A. Bender, Alex Conway, Martin Farach-Colton William Jannen, Rob Johnson, Donald E. Porter, and Jun Yuan. *FestiFS: A Compleat File System for Commodity SSDs*.

[FAST'20]: Yang Zhan, Alex Conway, Yizheng Jiao, Nirjhar Mukherjee, Ian Groombridge, Michael A. Bender, Martin Farach-Colton William Jannen, Rob Johnson, Donald E. Porter, and Jun Yuan. *How to Copy Files*.

[ICPR'19]:Wenhui Zhang, **Yizheng Jiao**, Dazhong Wu, Srivatsa Srinivasa, Asmit De, Swaroop Ghosh, Peng Liu. *ArmorPLC: Cyber Security Threats Detection through Ladder Logic Validation for PLCs*.

[HotStorage'19]: Alex Conway, Eric Knorr, **Yizheng Jiao**, Michael A. Bender, William Jannen, Rob Johnson, Donald Porter, Martin Farach-Colton. *Filesystem Aging: It's more Usage than Fullness*.

[SPAA'19]:Michael A. Bender, Alexander Conway, Martin Farach-Colton, William Jannen, **Yizheng Jiao**, Rob Johnson, Eric Knorr, Sara McAllister, Nirjhar Mukherjee, Prashant Pandey, Donald E. Porter, Jun Yuan and Yang Zhan. *Small Refinements to the DAM Can Have Big Consequences for Data-Structure Design*.

[FAST'18]:Yang Zhan, Alex Conway, **Yizheng Jiao**, Eric Knorr, Michael A. Bender, Martin Farach-Colton, William Jannen, Rob Johnson, Donald E. Porter, Jun Yuan. *The Full Path of Full-Path Indexing*. Oakland, USA. Feb, 2018.

[FAST'17]:Alex Conway, Ainesh Bakshi, **Yizheng Jiao**, Yang Zhan, Michael A. Bender, William Jannen, Bradley C. Kuszmaul, Donald Porter, Jun Yuan, Martin Farach-Colton. *File System Fated for Senescence? Nonsense, Says Science!*. Santa Clara, USA. Feb, 2017.

[SOSP'15]:Chia-Che Tsai, Yang Zhan, Jayashree Reddy, **Yizheng Jiao**, Tao Zhang, Donald E. Porter. *How to Get More Value From Your File System Directory Cache*. Monterey, USA. Jan, 2015.

[FAST'15]:William Jannen, Jun Yuan, Yang Zhan, Amogh Akshintala, John Esmet, **Yizheng Jiao**, Ankur Mittal, Prashant Pandey, and Phaneendra Reddy, Leif Walsh, Michael Bender, Martin FarachColton, Rob Johnson, Bradley C. Kuszmaul, Donald E. Porter. *BetrFS: A Right-Optimized Write Optimized File System*. Santa Clara, USA. Jan, 2015.

[SC'13]:Xiaobing Li, Yandong Wang, **Yizheng Jiao**, Cong Xu, Weikuan Yu. *Coomr: Cross-Task Coordination for Efficient Data Management in MapReduce Programs*. Denver, USA. November, 2013.

[PACT'13]:Bin Wang, Bo Wu, Dong Li, Xipeng Shen, Weikuan Yu, **Yizheng Jiao**, Jeffrey S. Veter. *Exploring Hybrid Memory for GPU Energy Efficiency through Software-Hardware Co-Design*. Edinburgh, Scotland. September, 2013.

[MASCOT'13]:Bin Wang, **Yizheng Jiao**, Xipeng Shen, Dong Li, Weikuan Yu. *A Versatile Performance and Energy Simulation Tool for GPU global memory*. San Francisco, USA. August, 2013.

[BAWD'13]:Yandong Wang, **Yizheng Jiao**, Cong Xu, Xiaobing Li, Teng Wang Xinyu Yue, Cristi Cira, Bin Wang, Zhuo Liu, Bliss Bailey, Weikuan Yu. *Assessing the Performance Impact of HighSpeed Interconnects on MapReduce*. 2013.

TECHNOLOGY SKILLS

Programming Languages: C, Python, Java

Operating System and Services: Linux, AWS Cloud Services, Docker, Jenkins.

HONORS AWARDS

- Merit Student Scholarship of 2009, 2010, 2011;
- University-Wide Thesis Award of Chongqing University in 2009;
- National Aspiration Scholarship for two time in 2007 and 2009;
- Huawei Scholarship in 2006.